

Cse 312 HW 7

1) I have read the syllabus and agree to follow the collaboration policies

2) a) $\binom{12}{4}$ or $\boxed{495}$ order does not matter, therefore you just choose 4 out of the 12.

b) $\frac{12!}{8!}$ Here, order matters because the positions are distinct.

c) $\binom{12}{4} - \binom{7}{4}$ We take all possibilities of infidder positions, subtract all possibilities where at least 1 Faculty is not represented.

3) a) $\binom{240}{130}$ 240 total steps (130 x, 160 y) and you just choose which 130 steps of 240 come when so its 240 choose 130.

b) $\binom{160}{70} \cdot \binom{130}{60}$ I break into two steps, first 160 choose 70 to calculate paths from $(0,0) \rightarrow (70,40)$ then $\binom{130}{60}$ to go from $(70,40)$ to $(130,160)$

$$c) \binom{240}{130} - \left(\binom{70}{30} \binom{220}{100} + \binom{160}{70} \binom{130}{60} \right) - \left(\binom{70}{30} \binom{40}{10} \binom{130}{60} \right)$$

I find total paths, and then subtract the paths through $(30,40)$ and the paths through $(70,40)$. This over counts, as paths through both are subtracted twice, so I re-add paths through both.

4) $\binom{5}{1} \cdot \binom{11}{2} \cdot (11)^4$ Since there are 5 suits, but 6 cards in the hand, one has to appear twice, hence $\binom{5}{1}$. Then choose two cards from that suit $\binom{11}{2}$, then 1 card from the remaining 4 suits.

5 a) $\frac{10!}{2!}$ 10 seat options, you take out the two extra seats to avoid overcounting

b) 1 2 3 4 5 6 7 8 9 10
 1 2 3 4 5 6 7 8 9 10
 1 2 3 4 5 6 7 8 9 10
 1 2 3 4 5 6 7 8 9 10

$$\frac{7!}{2!}$$

7! comes from treating CD and EFG as one unit, and thus removing the number of free options similar to the couple seating from section

c) $\frac{7!}{2!} \cdot (2!)^2 \cdot 2$ This is just adding in the new possible arrangements (each is $(2!)$; CD, DC, EFG, GFE).

d) $\left(\frac{6!}{2!}\right) \cdot (2!)^3$ Removes another seating option, like part b; adds another multiply by 2, because AB can be in any order.

e) $\left(\frac{7!}{2!} \cdot 2^2\right) - \left(\frac{6!}{2!} \cdot (2!)^3\right)$ Subtracts the unwanted pairs (AB next to each other) from wanted pairings (complementary counting)

$$b) a) \sum_{i=0}^k \binom{k}{i} \binom{m}{i} = \binom{k+m}{k}$$

Imagine you are choosing k club members from a group of k returning members and m new applicants.

RHS: You take the total people trying to get into the club and choose k people from that group. The right side does this $\binom{k+m}{k}$ by adding returning and applicants and choosing k .

LHS: choose i people from the m applicants, $\binom{m}{i}$.

Then choose i old members of the club to exclude, which is done $\binom{k}{i}$ ways. Summing over all possible i values gives the LHS.

Since these ways both work, they must be equal.

$$b) \sum_{j=0}^y \binom{y}{j} \binom{x}{j} = \binom{y}{x} 2^{y-x}$$

Imagine choosing TAs from y applicants with a fixed amount of Leads x .

LHS: If you choose j TAs out of y applicants you get $\binom{y}{j}$ out of those j you must choose x lead TAs or $\binom{j}{x}$ summed based on the amount of applicants to get $\sum \binom{y}{j} \binom{j}{x}$.

RHS: choose x leads out of y applicants: $\binom{y}{x}$, then choose the remaining non-lead TA. There are two options: TA, or not, so the amount is therefore 2^{y-x} for the amount of applicants minus the amount already determined to be leads. Since these count the same thing, LHS = RHS.

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d) $\binom{120}{50} \binom{70}{40} \binom{30}{30}$ 120 total steps, choose 50 in the x direction, then choose 40 of the remaining 70 steps to be y, then the remaining 30 steps are z direction.

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$$c) \sum_{i=3}^{n-3} \binom{n-i}{3} \binom{i-1}{2} = \binom{n}{6} \quad n \geq 6$$

Suppose we are choosing 6 values from a set of n numbers, say $n = \{x_1, x_2, x_3, x_4, x_5, x_6\}$

By definition the RHS does this with $\binom{n}{6}$ or n choose 6.

The LHS also shows this. We can say that the set n is sorted into order, where i is the 3rd lowest value (minimum = 3 as seen in the summation). To find the values smaller than x_i , we say $(i-1)$ choose 2 or $\binom{i-1}{2}$ since there are chosen from the set $\{1, 2, \dots, i-1\}$. To find the other 3 values, we know that they are greater than x_i . Therefore they are taken from the set $\{i+1, i+2, \dots, n\}$. There are $n-i$ ways to represent this, so we say $n-i$ choose 3 or $\binom{n-i}{3}$. Multiplying together we get all 6 possibilities, the values lower than x_i and the 3 larger values. Since both sides take 6 numbers from n , they must be equal.

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b) The most confusing numpy Function/class to me would be the aggregate function. I understand somewhat its use in data analysis and statistical measurements, but I still don't fully know its implementation. I don't understand the aggregation of different axis/ what that does and why it's significant.